

Probability

Order the following events from *least likely* to *most likely*.

a A dice is to be thrown.

A The number showing will be even.

B The number showing will be larger than 4.

C The number showing will be less than 6.

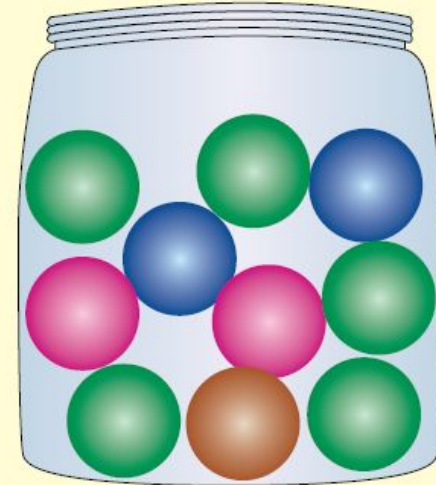


b A ball is to be drawn at random from the jar shown.

A The ball will be brown.

B The ball will be green.

C The ball will be blue or pink.

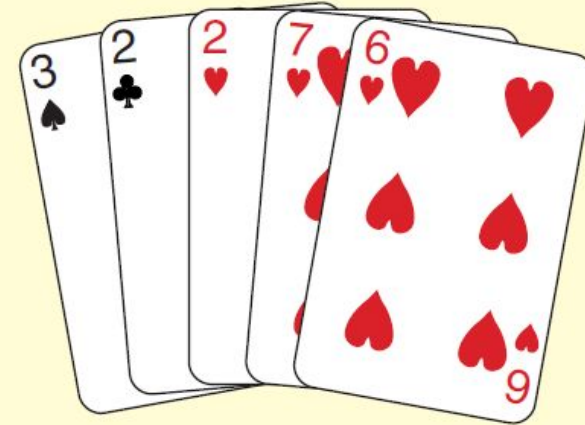


c One card is to be chosen at random from these five.

A The card will be a 2.

B The card will be a 7.

C The card will be greater than 3.



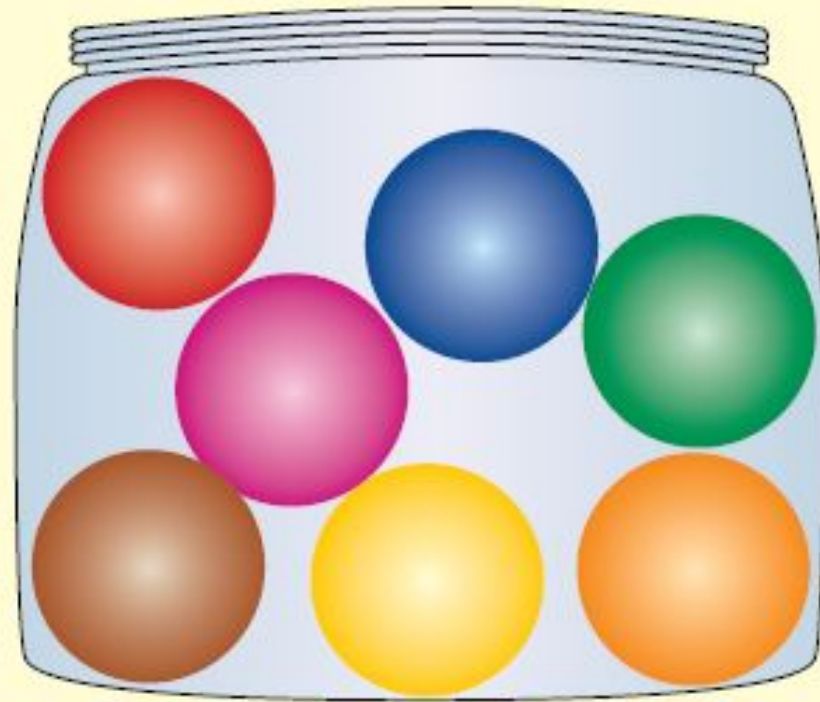
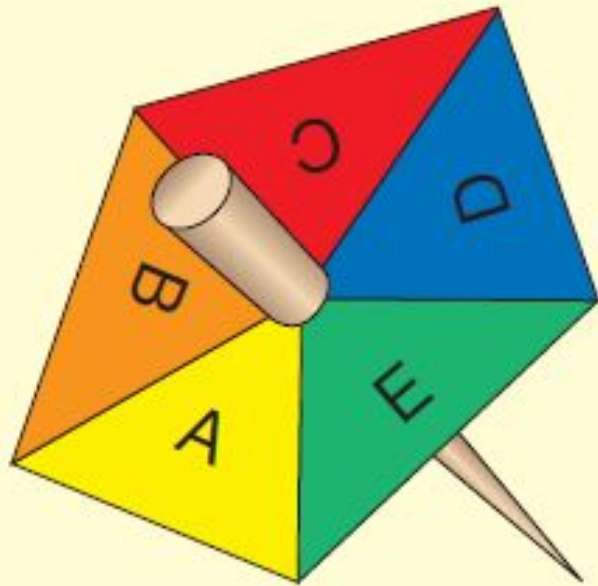
List all possible outcomes of each event.
That is, list the sample space.

a A dice is thrown.

b A coin is tossed.

c This spinner is spun.

d A coloured ball is chosen.



How many outcomes are in each sample space?

- a** A number between 4 and 7 is chosen at random.
- b** A ticket is chosen from tickets numbered 1 to 99.
- c** A ticket is chosen from tickets numbered 0 to 100.
- d** Ten cards are placed in a hat and one is chosen at random.

Liam throws a dice. Write, as a fraction, the probability that he will throw:

- | | |
|--------------------------------|-------------------------------|
| a a 2 | b a zero |
| c an even number | d an odd number |
| e either a 1 or a 6 | f a number less than 4 |
| g a number less than 7. | |



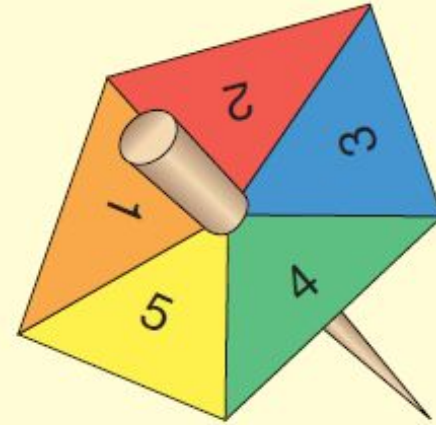
A coin is tossed.

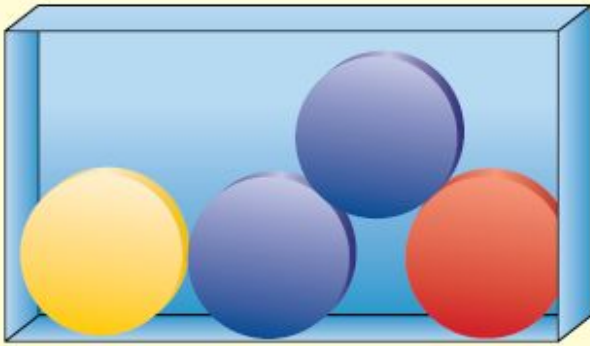
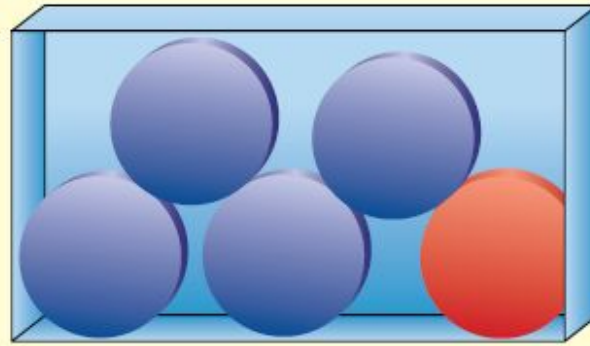
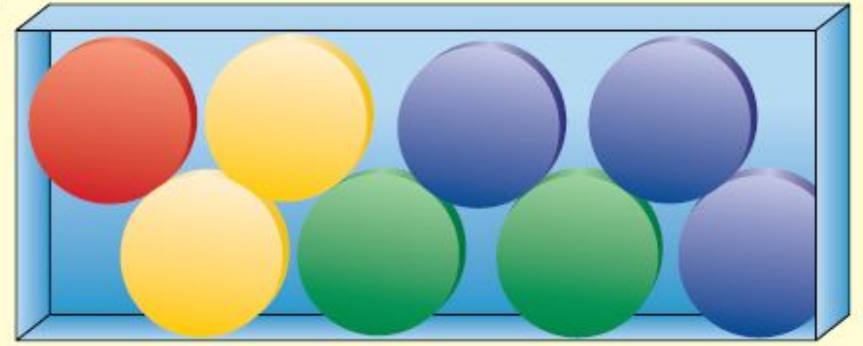
- a** Write, as a fraction, the probability of tossing:
 - i** a head
 - ii** a tail.
- b** Write as a percentage, the probability of tossing:
 - i** a head
 - ii** a tail.

The spinner to the right, marked 1 to 5 is spun.

Write, as a fraction, the probability of spinning:

- a** a 3
- b** an odd number
- c** an even number
- d** a zero.



A**B****C**

Counters have been slid into these narrow containers. One counter will be chosen at random from each container. For each container, find the probability of choosing:

- a** a red counter
- c** a yellow counter

- b** a blue counter
- d** a counter that is either red or blue.



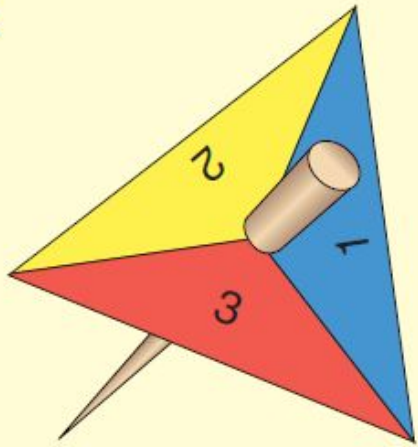
Of 100 tickets sold in a raffle, Luke bought 10, Heather bought 5 and Alan bought 1. A ticket was chosen at random to determine who won the prize. What is the probability that the prize was won by:

- a** Alan? **b** Luke? **c** Heather? **d** none of these three people?

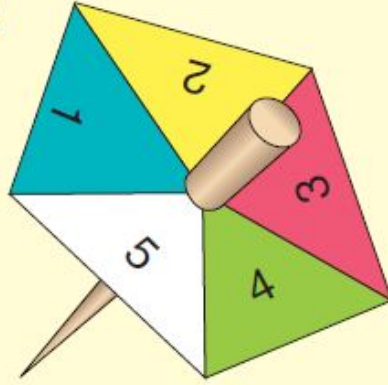
How many of the 100 tickets would I need to buy for the probability of my winning to be:

- e** 0? Explain why. **f** 1? Explain why.

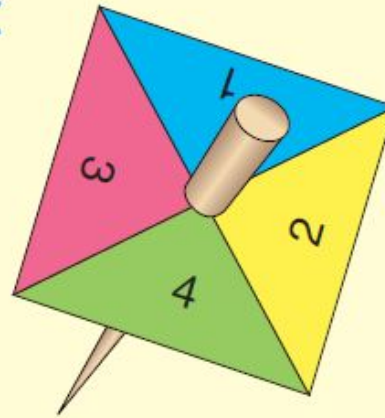
A



B



C

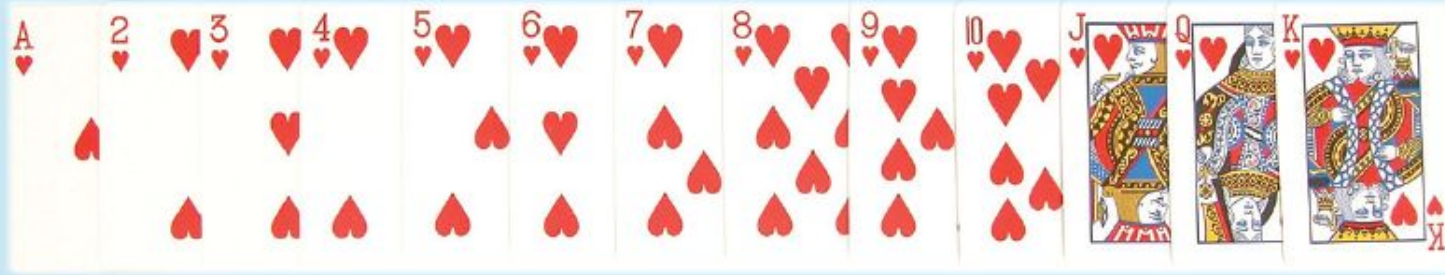


For which of these spinners is the probability of spinning a 3 equal to:

- a** 20% **b** 25% **c** $\frac{1}{3}$



A standard pack of cards has 4 suits: hearts (♥), diamonds (♦), clubs (♣) and spades (♠). There are 13 cards in each suit and 52 cards altogether.



The first of these cards is called an **Ace**. The last three cards are the *picture* or *court* cards: the **Jack**, the **Queen** and the **King**.

From the 13 cards shown above, one card is to be chosen at random. What is the probability that the card chosen will be:

- | | | |
|-------------------|-------------------|------------------------------|
| a the 7? | b the Ace? | c a picture card? |
| d a heart? | e a spade? | f either a 9 or a 10? |

From a standard pack of 52 cards, a card is chosen at random. What is the probability that the card will be:

- | | | |
|---------------------|--------------------------|-----------------------------------|
| a a diamond? | b a red card? | c a black card? |
| d a 3? | e a picture card? | f either an Ace or a King? |